

(Research) Reward and Observe Higher-Order Goalderivatives (2025)

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This repository researches into possible improvements to Goal-Oriented Reinforcement Learning (RL) by evaluating the **Differential Goalkinematic State (DGS)**, whose components are based on the distance to the goal - in the following called goal-directed derivatives or simply *goalderivatives*.

The probed improvements include sample-efficiency during training and generality of the resulting policy to unseen goals.



Desired Goal

A **goalderivative** $d^{(k)}$ of order $k > 1$ is the rate of change in the scalar distance d (specific to e.g. the L^2 -norm) or its derivatives (velocity $d^{(1)}$, acceleration $d^{(2)}$, jerk $d^{(3)}$ etc.) *towards* a numerically defined goal.

In time-discrete environments, the goal derivatives at a time t_i can be recursively estimated with the distance to the goal (*goaldistance*) at t_i by backward difference:

$$d^{(0)}(t_i) = d(t_i) := \text{goaldistance at time } t_i$$

$$d^{(k)}(t_i) \approx \frac{d^{(k-1)}(t_i) - d^{(k-1)}(t_{i-1})}{t_i - t_{i-1}}$$

In a vector, multiple goal derivatives of $k > 1$ at t_i form a differential kinematic state vector

$$s_{DGS}(t_i) = \begin{pmatrix} d^{(1)} \\ d^{(2)} \\ \vdots \\ d^{(k)} \end{pmatrix} (t_i)$$

The idea is now to evaluate the vector for either reward design, observation augmentation, or both.

Summary

Instead of using only the distance to the goal, its derivatives are also considered - in the reward function or as part of the observation in RL.

RL algorithms are applied to a formal **Markov Decision Process (MDP)**

$$\begin{aligned} M &= (\text{states, actions, transition probabilities, discountfactor, rewards}) \\ &= (S, A, T, \gamma, R) \end{aligned}$$

to find, for each **state** $s_k \in S$, the optimal **action** $a_k^* \in A$ that maximizes the expected γ - discounted return

$$G = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]$$

of experienced **rewards** $\{r', r'', \dots\} \subseteq R$ from its next states $\{s', s'', \dots\} \subseteq S$, which are reached with probabilities in T . However with RL, T and R are initially unknown and must be gradually discovered by exploration similar to *trial and error*. [1]

The result is an optimal **policy** $\pi^* : S \rightarrow A$ that assigns to each state its optimal action. Note that a higher-level goal is not formalized in the MDP - its optimality does not ensure the objective success in reaching (or keeping) an indirect goal. Careful and effective design of the underlying MDP is therefore crucial for the correct and efficient convergence of RL.

Research Claim 1 (Goalderivative Potential-Based Reward Shaping)

In goal-oriented RL training towards an optimal policy, by adding to the existent rewards a potential-based shaping term based on the DGS vector, the training can be more efficient without changing the original optimal policy.

In this work, rewards are deterministic, i.e. they are assigned to states by the real **reward function** $R : S \times A \rightarrow \mathbb{R}$, meaning at state s_t the algorithm receives a reward $r_t = R(s_t, a_t)$. It is proven that by adding a strictly **shaping function** $F(s_t, a_t) = \gamma\Phi(s_{t+1}) - \Phi(s_t)$ with state-dependent potentials Φ , the reward function can be modified to

$$R' = R + F$$

without changing the optimal policy: By solving the modified MDP $M' = (S, A, T, \gamma, R')$ we also solve the original MDP M .

With a naive goalkinematic potential

$$\Phi_{naive}(s) = -\left\| \begin{pmatrix} d \\ s_{DGS} \end{pmatrix} \right\|_2 = -\left\| \begin{pmatrix} d \\ d^{(1)} \\ d^{(2)} \\ \vdots \\ d^{(k-1)} \end{pmatrix} \right\|_2,$$

achieving states that are not only close, but are also expected to be closer in the next states, will be rewarded higher - their goalderivatives are more favorable. On the other hand, achieving states that are only statically close, or even moving away, will be rewarded lower. However, such a frequent rewarding enables *reward hacking*, where the agent might oscillate between moving closer and farther from the goal without actually reaching the goal, i.e. the converged policy is not the optimal policy, let alone a successful one. With a **conjunctive goalderivative potential (CGP)**

$$\Phi_{conj}(s) = \begin{cases} 1, & \text{if } \forall k : d^{(k)} < 0, \\ -1, & \text{if } \forall k : d^{(k)} > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (d^{(k)} \in s_{DGS})$$

that type of reward hacking is less probable with higher order k , since isolated goalderivatives are not rewarded anymore. Since they are also not penalized, exploration is allowed - this is especially beneficial in multi-goal environments.

In 5 simulations à 50 epochs (one epoch consists of 200 episodes à 50 timesteps) within Gymnasium's sparse *HandReach* environment [2], where a robotic hand is trained to reach different

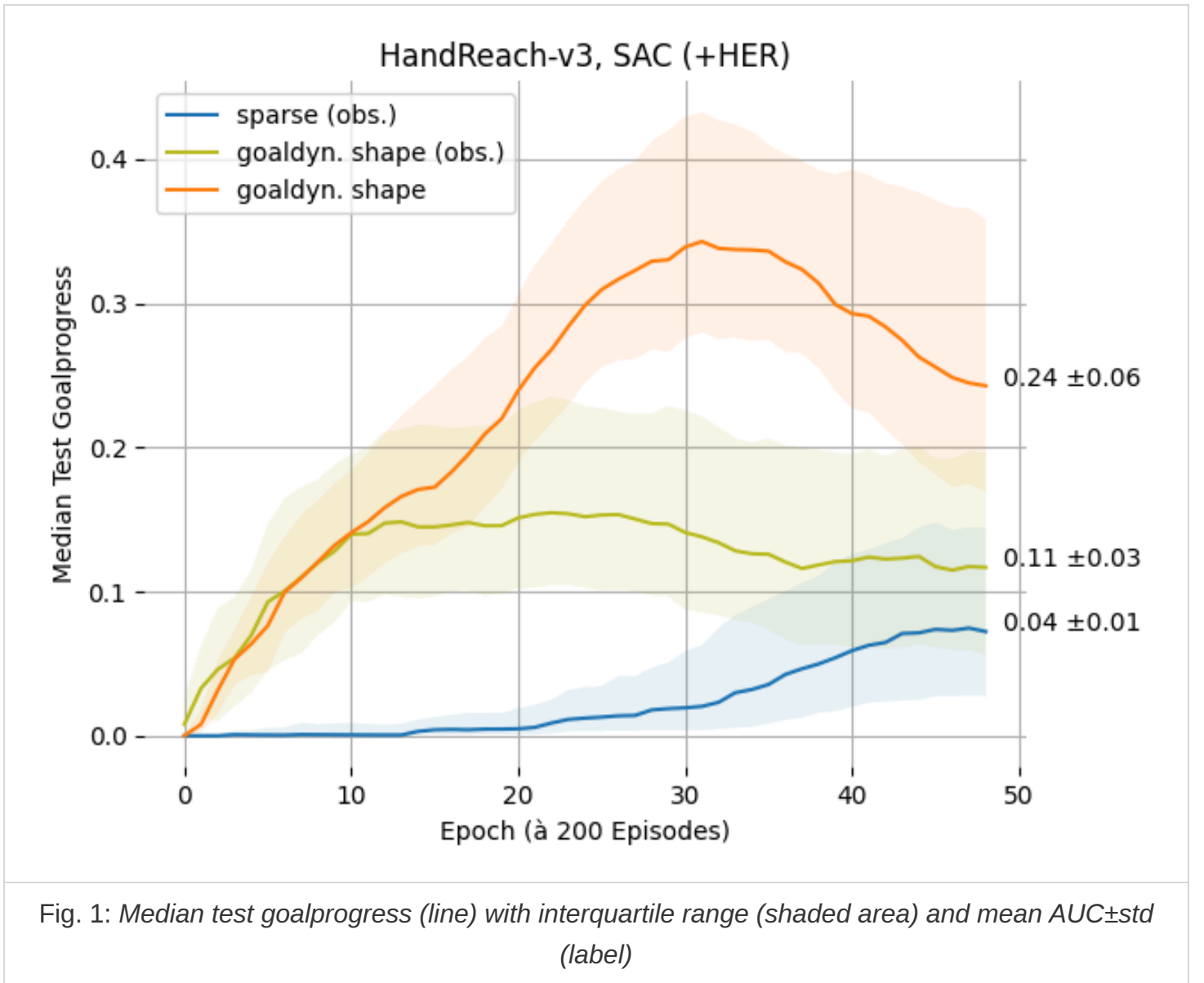
coordinates with its fingertips, the shaped reward ($k = 3, \gamma = 0.95$)

$$\begin{aligned}\hat{R}_1(s, a) &= R(s, a) + F(s, a) \\ &= R(s, a) + 0.95 * \Phi_{conj}(s') - \Phi_{conj}(s)\end{aligned}$$

increased the training **sample efficiency**, defined as the area under the curve (AUC) of the test *goalprogress*

$$P = \frac{d_0 - d_{end}}{d_0} \quad (0 \leq P \leq 1) (d_{end} := \text{goaldistance at the end of the episode}),$$

by factor **2.75** (Fig. 1: yellow line) compared to the unshaped baseline reward (Fig. 1: blue line) (median). The details of the experiments are described in the full work.



Note: With potential-based shaping, this claim theoretically requires that the goaldistance and the DGS are part of the observable state space, which is a separate focus in this research. In the experiments,

the efficiency gains with non-observable goal distance and DGS were even more substantial with factor 6 (Fig. 1: orange line), i.e. possibly justifying the theoretical violation of the Markov assumption.

Research Claim 2 (Goal derivative Reward Design)

In goal-oriented RL training towards an optimal policy, by designing rewards based on the DGS vector, the training can be more efficient.

(Formal Proof)

(Experimental Proof) (fetchpush)

Research Claim 3 (Goal kinematic Observation Augmentation)

In goal-oriented RL training towards an optimal policy, by adding goal kinematic information to the observation space, the training can be more efficient.

(adding to the Markov property)

(Formal Proof?)

(Experimental Proof) (fetchpush)

Research Claim 4 (Goal kinematic Observation Reduction)

In goal-oriented RL training towards an optimal policy, by reducing the observation space to contain *only* goal kinematic information, the training can be successful, more efficient and more general.

define success (vs. goal progress) define general

(Experimental Proof) (fetchpush)

Case Study: Fluent Visual Imitation of Hand Gestures by a Robotic Hand (Multi-Goal RL with Multi-Dimensional Goals)

(reward function granularity)

(hand gesture imitation)

Hyperparams

Hyperparams	Relevance	Examples	Comment
γ (discount)	+++	0.5 (shortsight: non-term., direct, fast/dumb), 0.95 (longsight: term., indirect, prudent/careful)	watch actor NN learning loss
shaping vs. redesign	++		
goalkin. reward aggregate	+++	all(DGS) (term.), any(DGS) (non-term.)	
goalderiv. order k	++	oscillation-res. (>2)	
goalkin. obs.	+++	deltas, dist/derivs., augm. vs reduce	
goalobs.	++	single-goal vs. multi-goal	
non-lin. activ.	+	ReLU (pos.) vs. Tanh (neg., normalized goal)	
reward freq.	+++	sparse (indirect) vs. dense (direct, straight)	

Anecdotes

“with enough goaldimensions, the goaldistance becomes meaningful goalprogress”

“efficiency does not necessarily mean success/effectiveness”

TODO

- hyperparams
- overviewing tables
- different algo(s)
- different env(s)
- “proving” imgs
- “proving” graphs (eg. training process metric?)
- “proving” gifs (eg. sped up training process video?)
- disclaimer (autonomy)
- license
- dense HER baseline, too?
- goalprogress vs. success
- (compare/combine with HER, vs. (normalized) multi-dim. goals w/o threshold, ie. inexact goals with different reachable (unknown) thresholds, adaptability/generality to unseen goals (interpolative vs. extrapolative))
- C2,C3 w/o HER
- C4 w/ HER
- sparse baseline unobs.

References

- HER
- HER envs.
- RL bible (barto, sutton)
- MDP
- markov property
- goaldistance
- potential-based shaping
- reward engineering
- observation augmentation
- observation reduction (for generality)
- multi-goal
- <https://openai.com/index/ingredients-for-robotics-research/>
- https://wandb.ai/rodrigodelazcano/gym_robotics/runs/1s3fuwye?nw=nwuserrodrigodelazcano

[1]

A.G. Barto, Reinforcement learning: An introduction. By richard’s sutton, SIAM Rev 6 (2021) 423.

[2]

M. Plappert, M. Andrychowicz, A. Ray, B. McGrew, B. Baker, G. Powell, J. Schneider, J. Tobin, M. Chociej, P. Welinder, others, Multi-goal reinforcement learning: Challenging robotics environments and request for research, arXiv Preprint arXiv:1802.09464 (2018).